

Physical Chemistry (I)

Lecture 13

Phases of Pure Substances



In the Last Class

- Classical statistical mechanics
 - Phase space
 - Calculation of partition function
- Applications
 - Perfect gas
 - Real gas
 - Equipartition theorem

Phase Space

Space of **all possible microstates** of a system

- Classical mechanics:

Space of all possible values of
position and **momentum** variables

Last class

Classical Partition Function

$$z = \sum_s e^{-E(s)/k_B T} \rightarrow z = \frac{1}{h} \int \int e^{-E(x,p)/k_B T} dx dp$$

for a 1-dimensional single particle

Calculation of Z

For indistinguishable particles in 3 dimensions,

$$Z = \frac{1}{N! h^{3N}} \left[\int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p} \right]^N = \frac{1}{N!} z^N$$

where

$$z = \frac{1}{h^3} \int \int e^{-E(\vec{r}, \vec{p})/k_B T} d\vec{r} d\vec{p}$$

Perfect Gas

- No interactions (no potential energy)
- Total energy = kinetic energy ($p^2/2m$)
- Bound in a box of volume V
 - Range of $x = (0, a)$
 - Range of $y = (0, b)$
 - Range of $z = (0, c)$
 - $abc = V$
- Range of momentum = $(-\infty, \infty)$

Last class

Z of Perfect Gas

$$z = V \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2}$$

For indistinguishable gas particles,

$$Z = \frac{1}{N!} z^N = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

Last class

Thermodynamics

$$Z = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2}$$

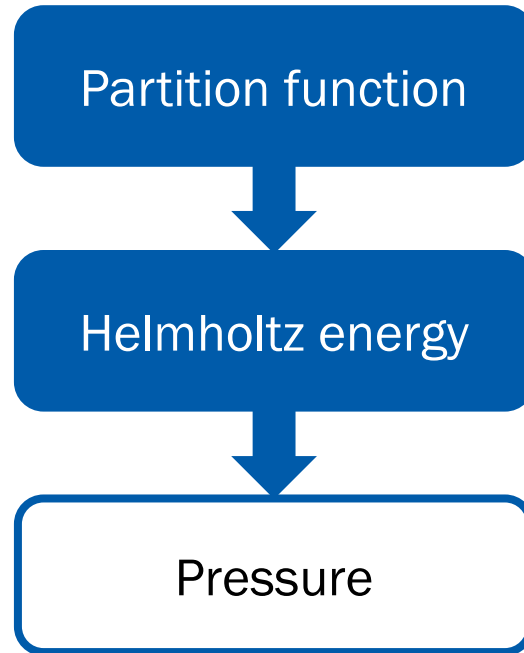
$$A = -Nk_B T \ln V + k_B T \ln N! - \frac{3}{2} Nk_B T \ln \left(\frac{2\pi m k_B T}{h^2} \right)$$

$$U = \frac{3}{2} Nk_B T$$

$$p = \frac{Nk_B T}{V}$$



Pressure of Real Gas



$$p = \frac{nRT}{V - nb} - a \frac{n^2}{V^2}$$

Last class

Quadratic Contributions

$$Z = \frac{1}{N! h^{3N}} \int \cdots \int e^{-E_{\text{tot}}/k_B T} d\vec{r}_1 d\vec{p}_1 d\vec{r}_2 d\vec{p}_2 \cdots d\vec{r}_N d\vec{p}_N$$

Assume that

$$E_{\text{tot}} = \sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2$$
$$(A_i, B_j > 0)$$

and that the ranges of both positions and momenta are $(-\infty, \infty)$

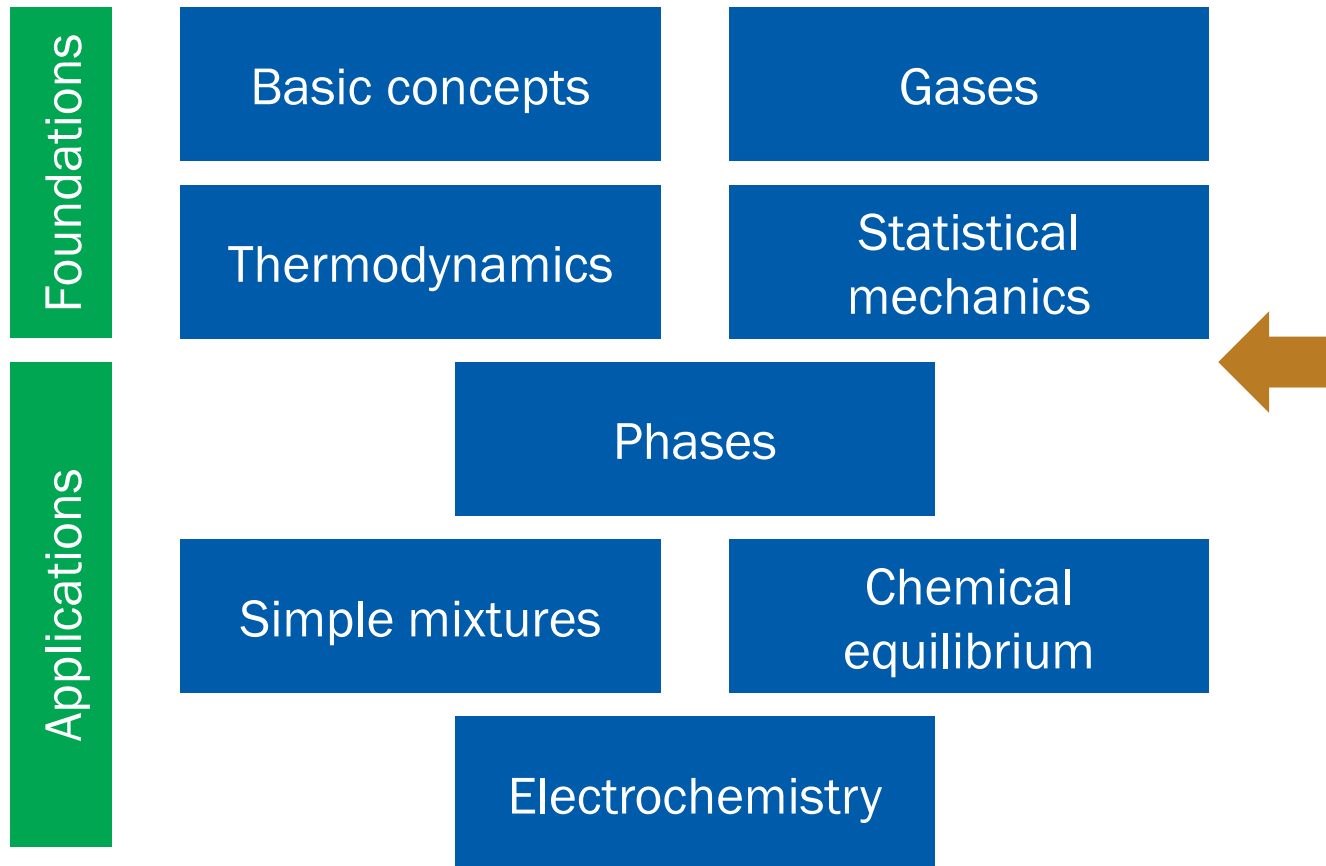
Equipartition Theorem

$$Z = \frac{\pi^{(f_r+f_p)/2}}{N! h^{3N}} \left(\frac{1}{\beta}\right)^{(f_r+f_p)/2} \prod_{i=1}^{f_r} \left(\frac{1}{A_i}\right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{1}{B_j}\right)^{1/2}$$

$$U = (f_r + f_p) \frac{k_B T}{2}$$

“The average value of each **quadratic contribution** to the internal energy is $(1/2)k_B T$.”

Where Are We?



7. Phase Transitions

Chapter Overview

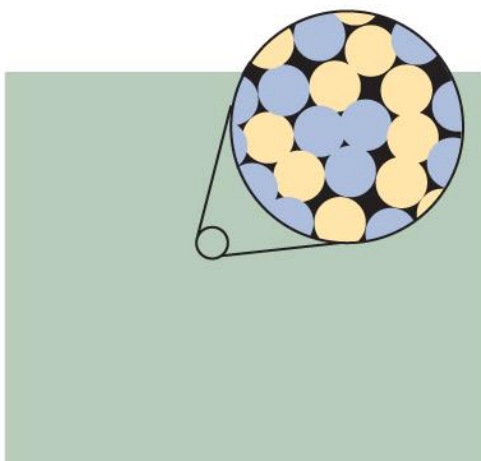
- Phase and chemical potential
 - Phase and phase diagram
 - Chemical potential
 - Phase rule
- van der Waals fluid
 - Liquid-gas phase transition
 - Supercritical fluid
 - Critical point and the principle of corresponding states
- Thermodynamic aspects of phase transitions



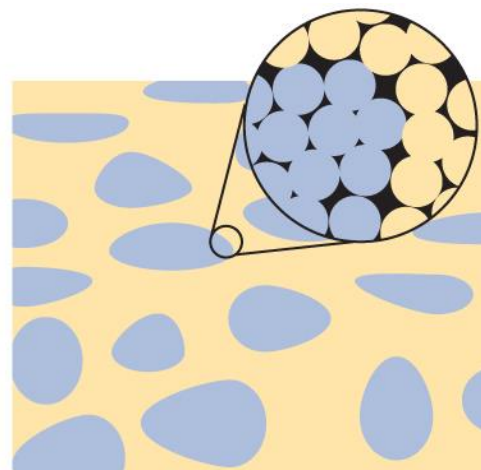
Phase

A form of matter that is **uniform** throughout
in **chemical composition** and **physical state**

Solution: Single Phase



solution
(one phase)



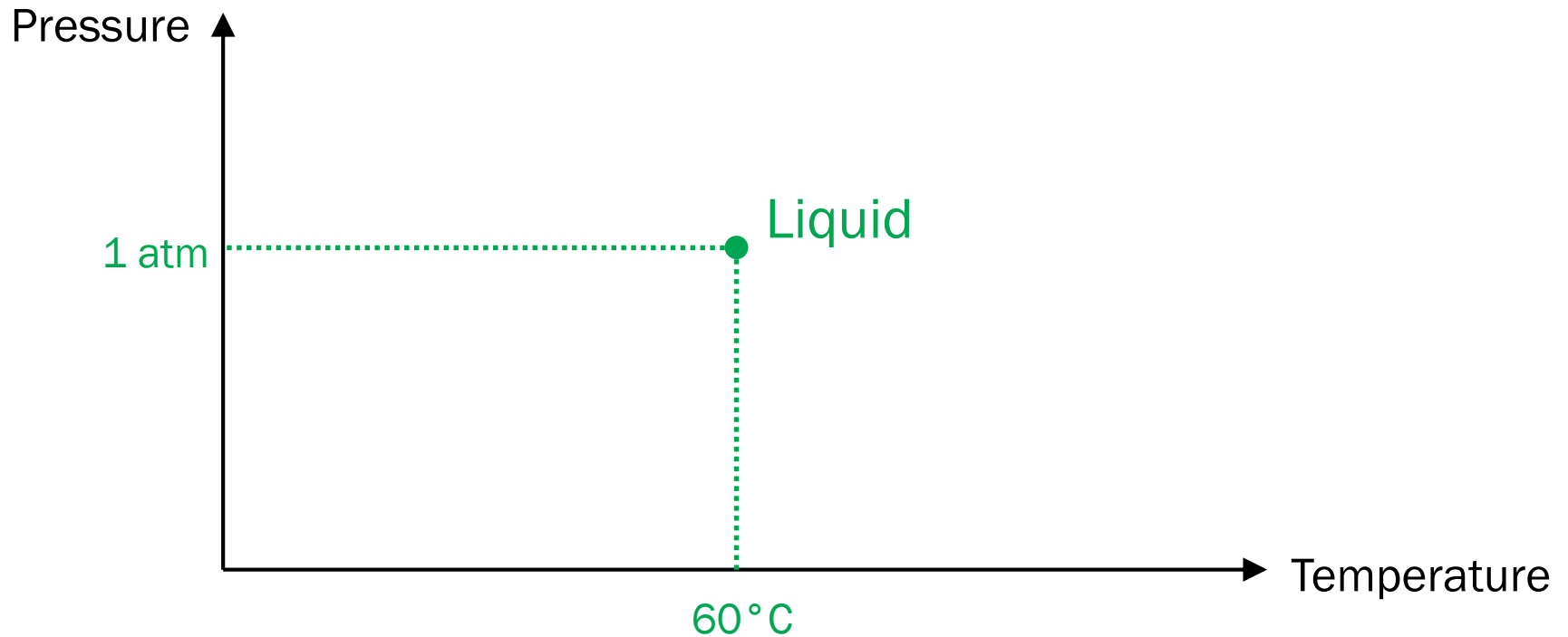
dispersion
(two phases)

Phase Diagram

A diagram that shows the regions at which its various phases are thermodynamically stable

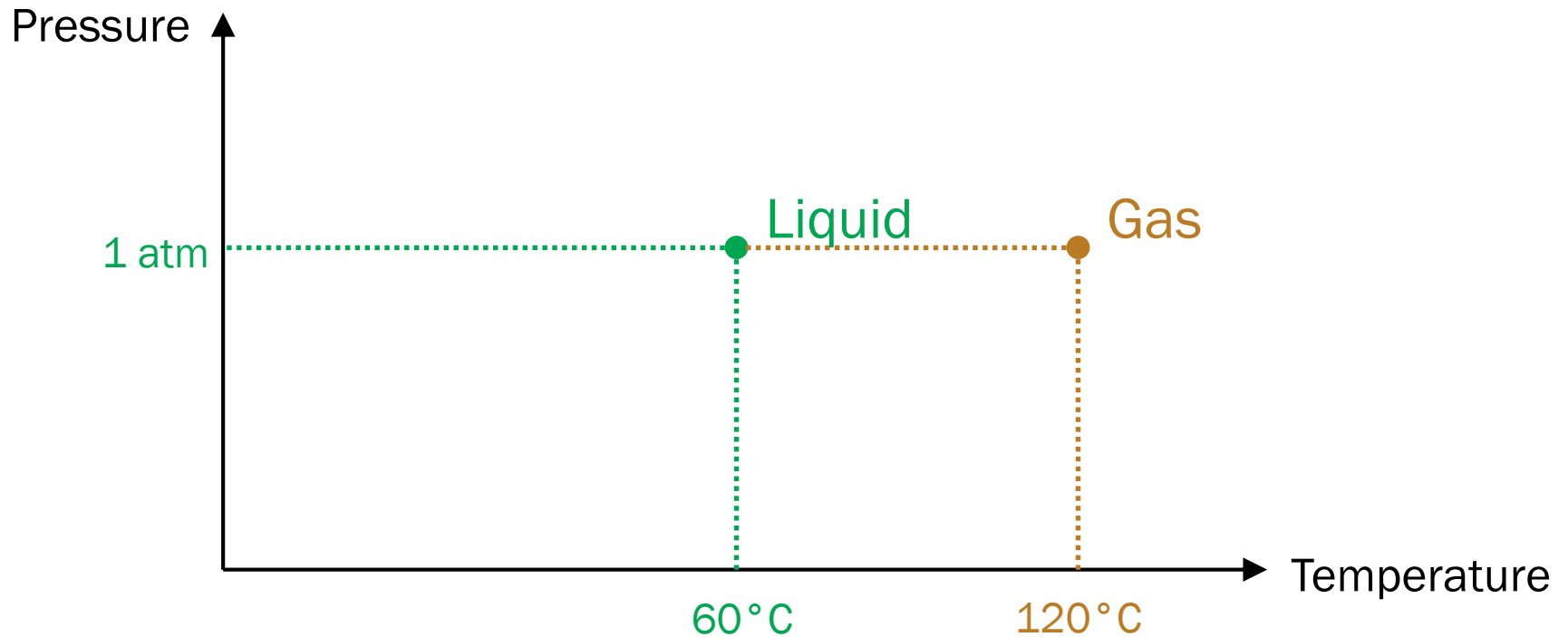
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Drawing Phase Diagram



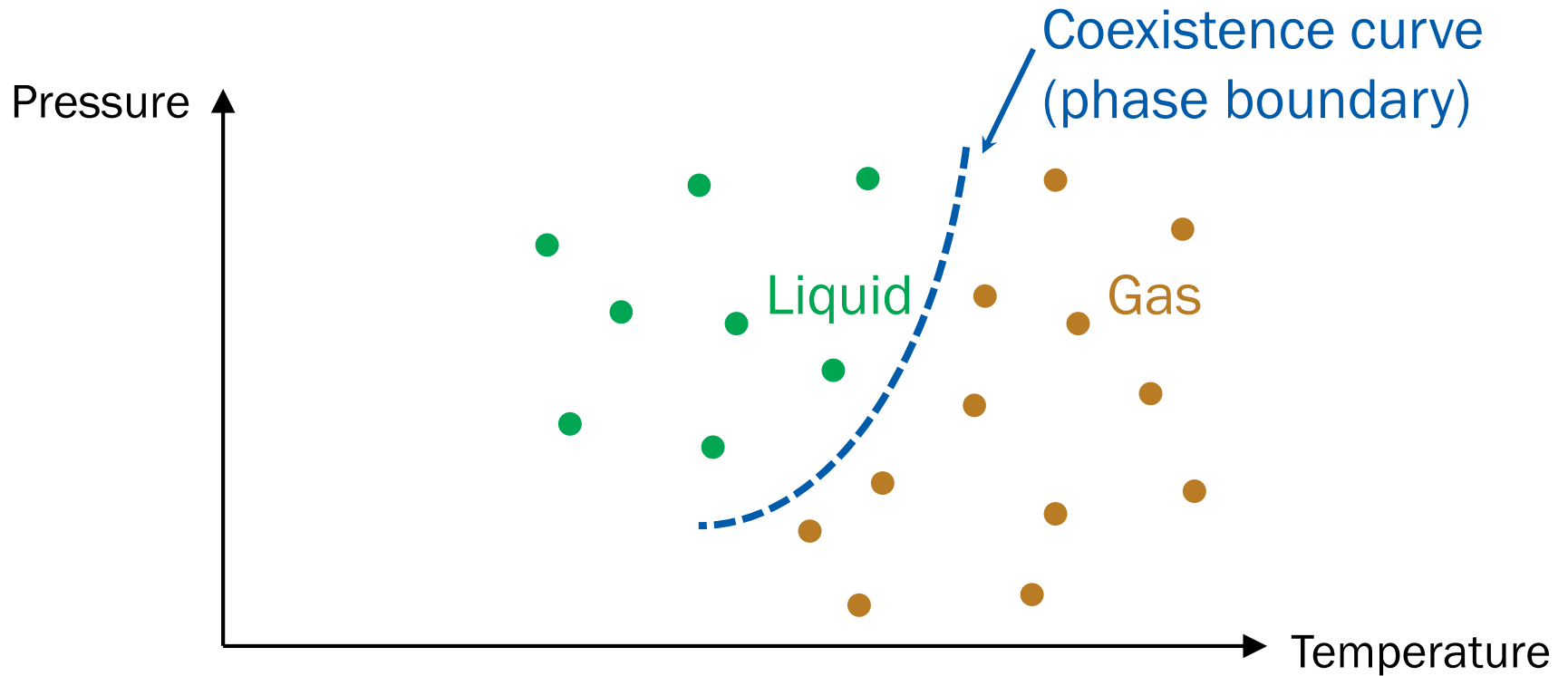
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Drawing Phase Diagram

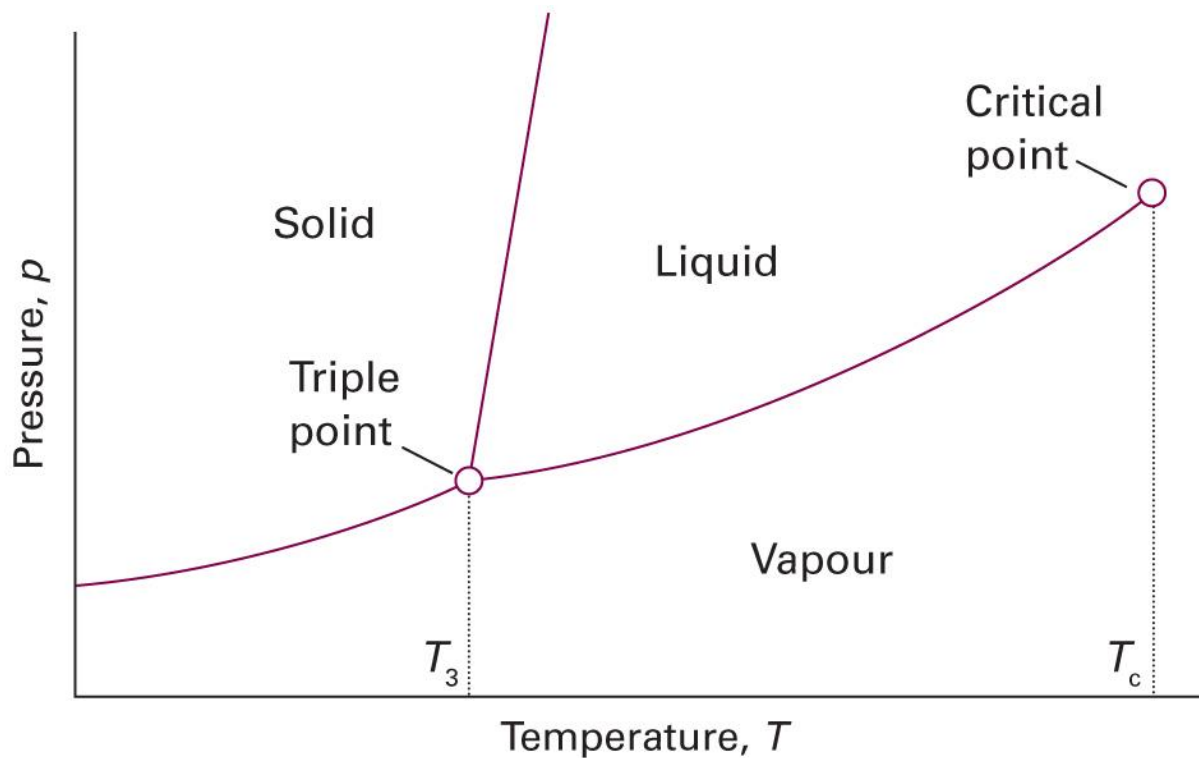


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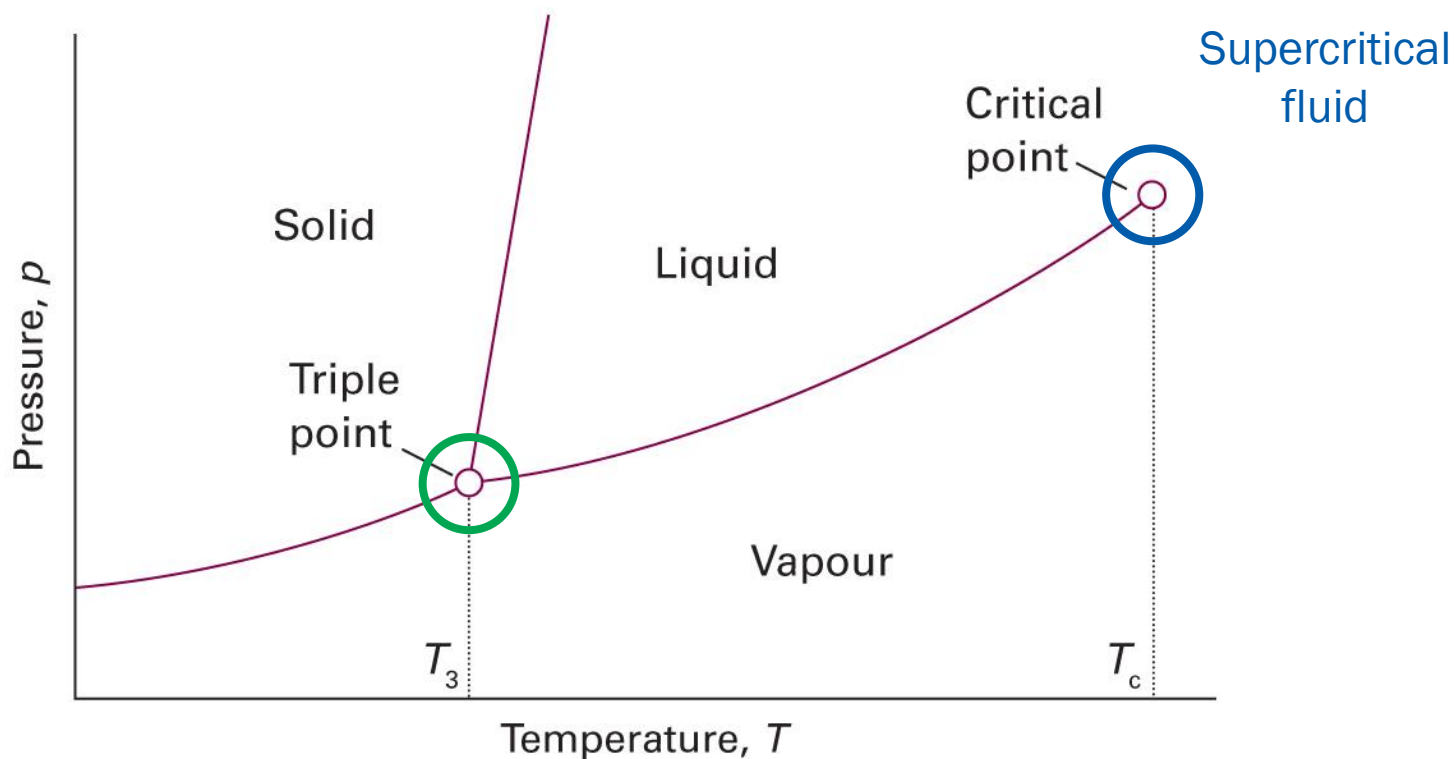
Drawing Phase Diagram



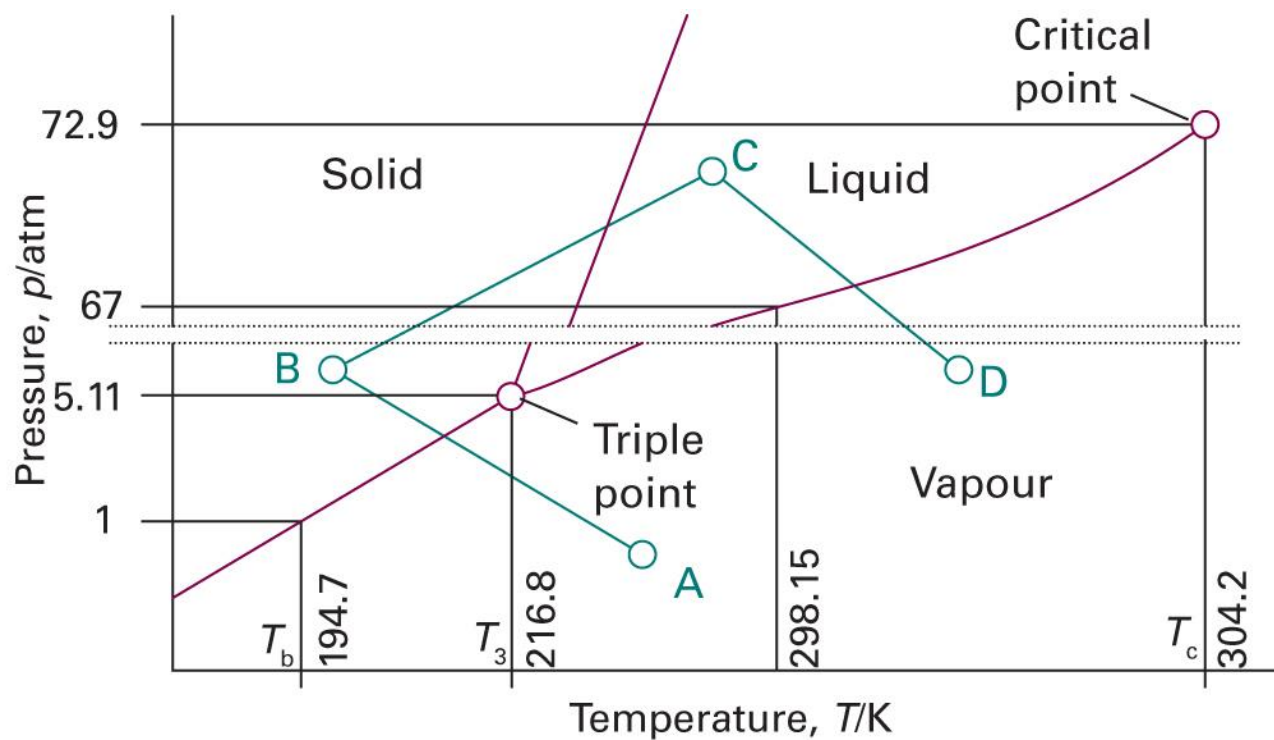
Typical Phase Diagram



Two Characteristic Points

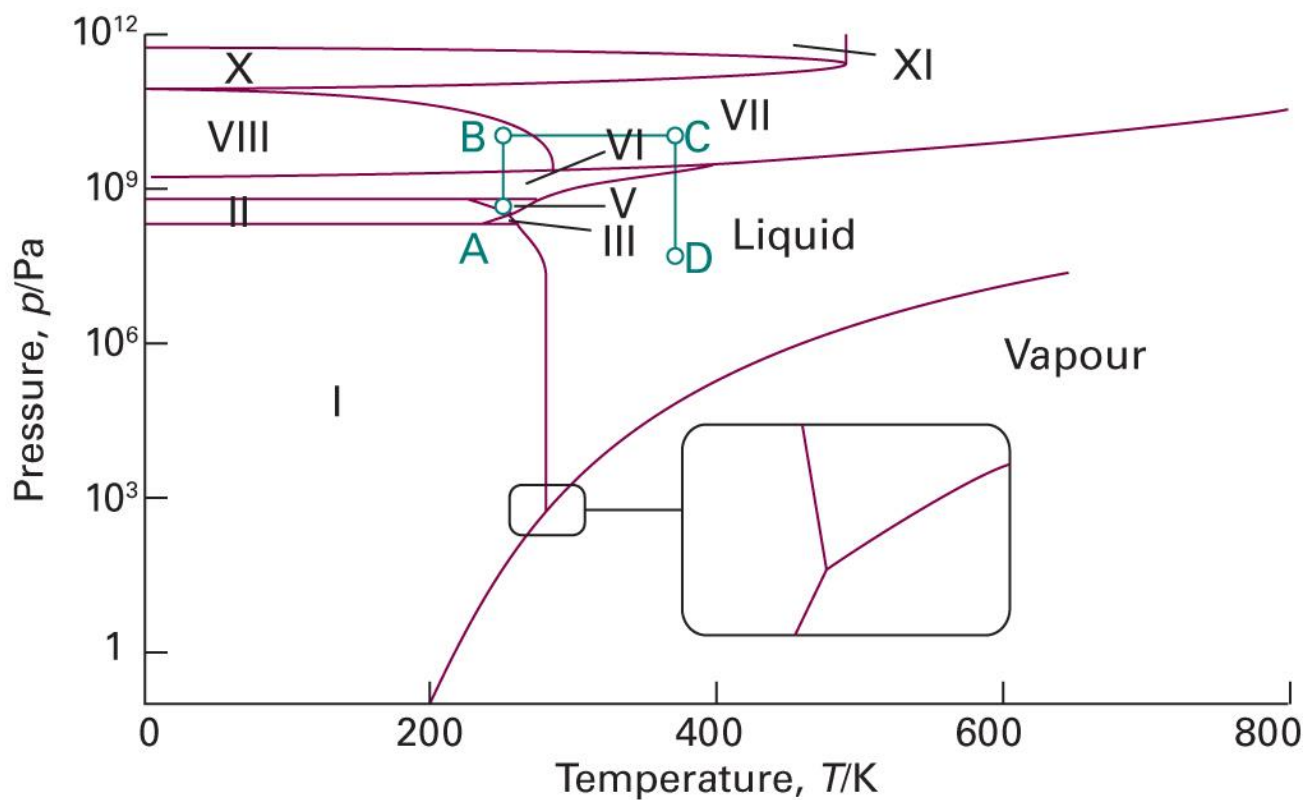


Carbon Dioxide



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figures 4A.8)

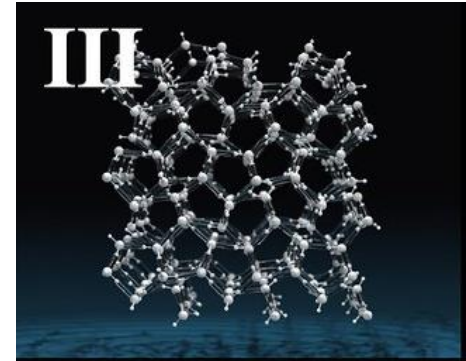
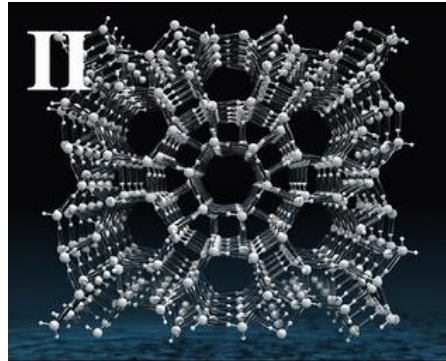
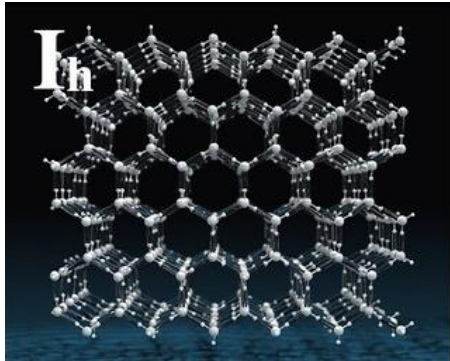
Water



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figures 4A.9)

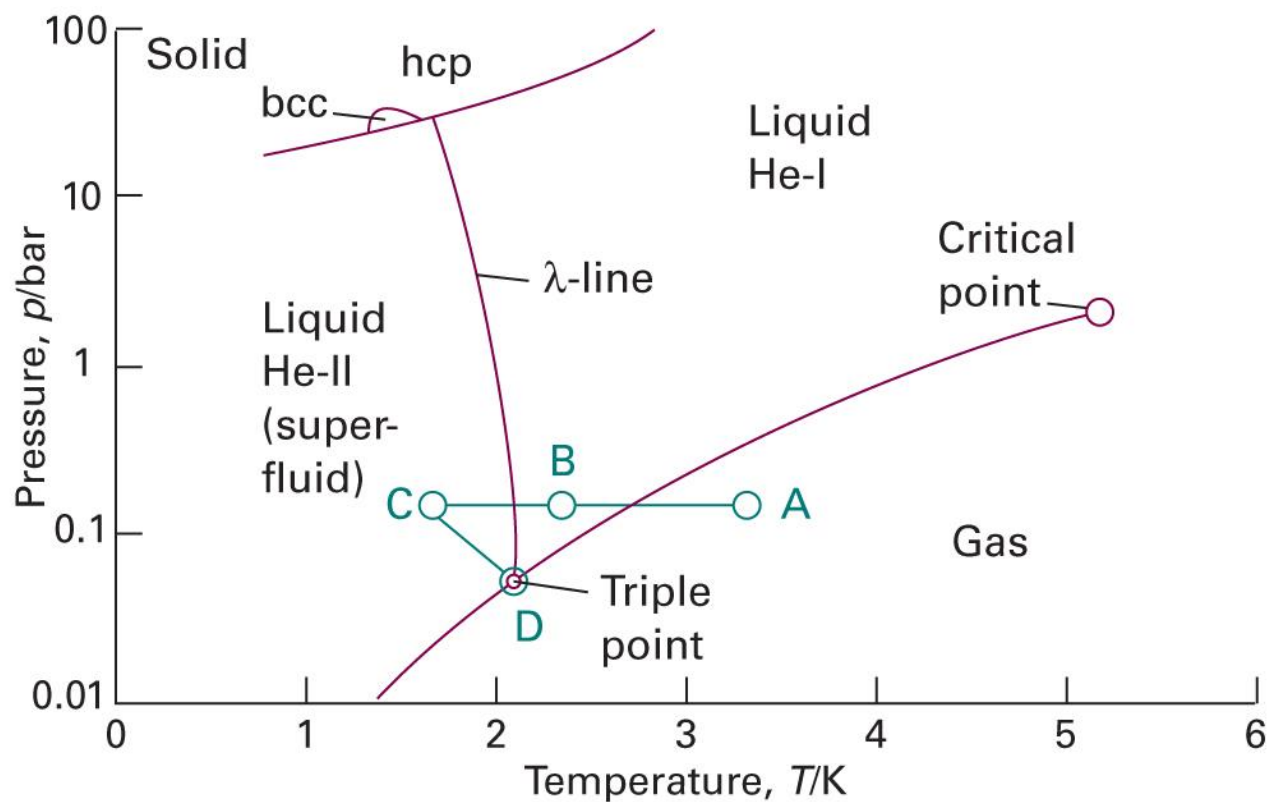
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Different Phases of Ice



<https://www.youtube.com/watch?v=UFGiBKYxHlw>

Helium (^4He)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figures 4A.11)

Phase Rule

“Can we have a quadruple point?”

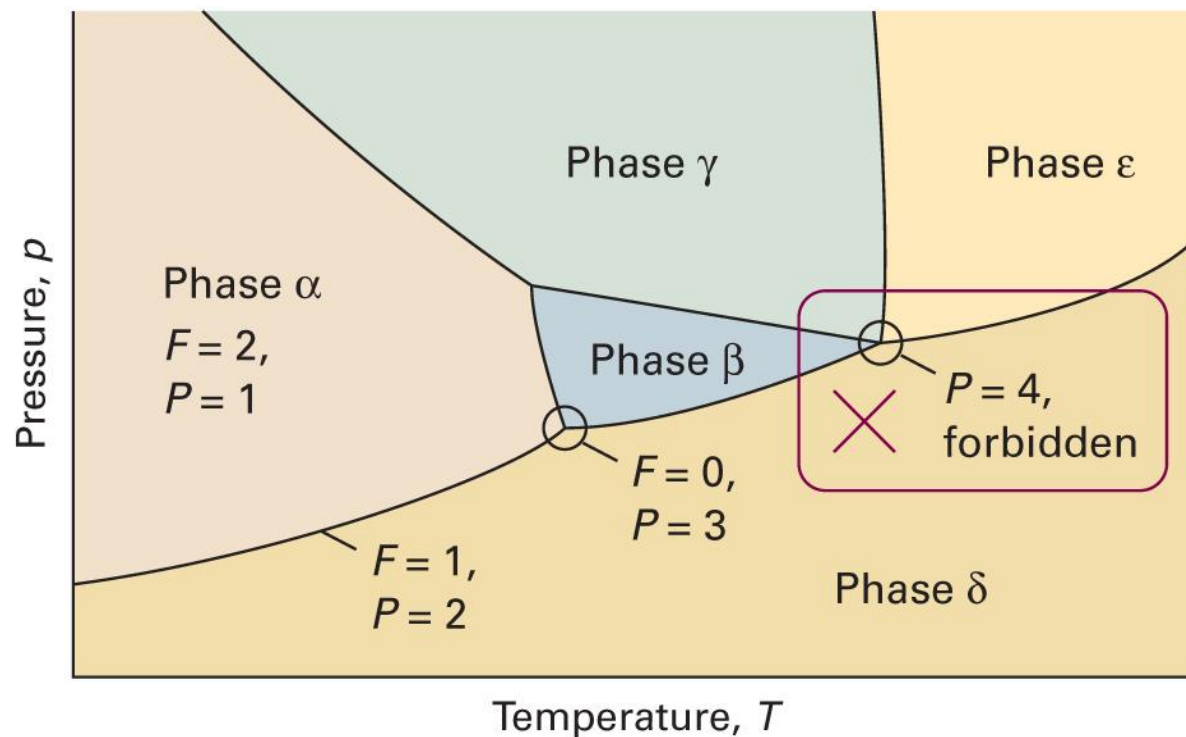
- **Phase rule:** a general relation between
 - Number of degrees of freedom F
 - Number of components C
 - Number of phases at equilibrium P
- Use the **criteria for phase stability** (same μ)
 - Recall that $\mu = g(T, p)$

Single Component ($C = 1$)

- $P = 1$: both T and p can be independently varied
 $\therefore F = 2$ (bivariant)
- $P = 2$: $\mu_1(T, p) = \mu_2(T, p)$
 $\therefore F = 1$ (univariant)
- $P = 3$: $\mu_1(T, p) = \mu_2(T, p)$ and $\mu_2(T, p) = \mu_3(T, p)$
 $\therefore F = 0$ (invariant)
- $P = 4$: two variables and three restrictions $\rightarrow$ impossible

In summary, $F = 3 - P$

Single Component ($C = 1$)



General Phase Rule

In general, the following rule holds:

$$F = C - P + 2$$

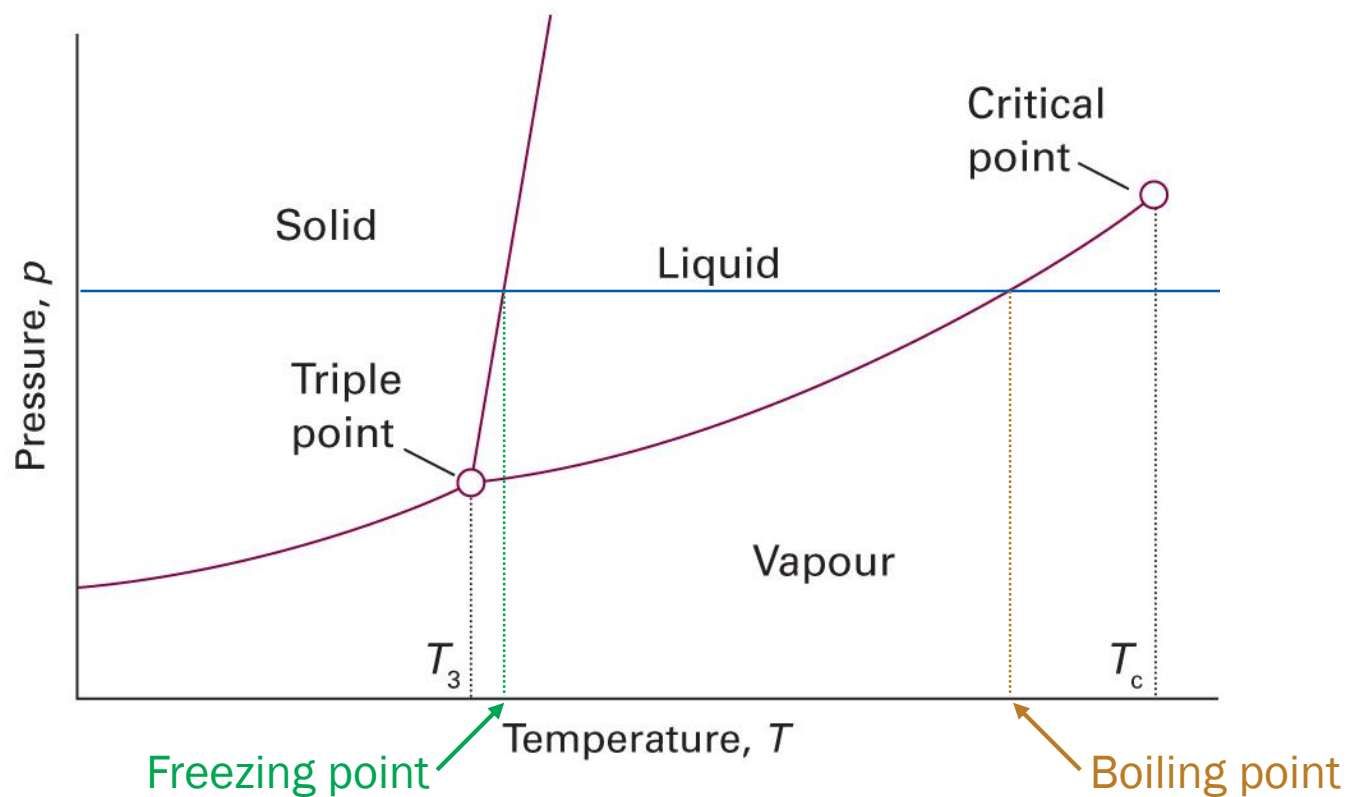
(Will be discussed in Chapter 7)

Phase Transition

Spontaneous conversion
of one phase into another phase

- occurs at transition temperature T_{trs} (for given p)
- At T_{trs} , the two phases are at equilibrium

Freezing and Boiling Points



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figures 4A.4)

Freezing and Boiling Points

- Normal boiling point T_b : b.p. at 1 atm
- Standard boiling point: b.p. at 1 bar
- e.g. water: $T_b = 100^\circ\text{C}$, standard b.p. = 99.6°C

- Normal freezing point T_f : f.p. at 1 atm
- Standard freezing point: f.p. at 1 bar
- Usually the difference between the two is negligible

Two Phases at Equilibrium

$$dU = TdS - pdV$$

- Mechanistic equilibrium: $p_1 = p_2$
 - Conjugate extensive variable: dV
- Thermal equilibrium: $T_1 = T_2$
 - Conjugate extensive variable: $dS = \delta q/T$
- Diffusive equilibrium: ?
 - Conjugate extensive variable: dn

Chemical Potential

- Diffusive equilibrium: $\mu_1 = \mu_2$
 - Conjugate extensive variable: dn
- (Modified) differentials of thermodynamic potentials

$$dU = TdS - pdV + \mu dn$$

$$dH = TdS + Vdp + \mu dn$$

$$dA = -SdT - pdV + \mu dn$$

$$dG = -SdT + Vdp + \mu dn$$

3 Independent Variables

$$dU = TdS - pdV + \mu dn$$

This implies that $U = U(S, V, n)$, which leads to

$$dU = \left(\frac{\partial U}{\partial S}\right)_{V,n} dS + \left(\frac{\partial U}{\partial V}\right)_{S,n} dV + \left(\frac{\partial U}{\partial n}\right)_{S,V} dn$$

Comparison gives

$$\left(\frac{\partial U}{\partial S}\right)_{V,n} = T, \quad \left(\frac{\partial U}{\partial V}\right)_{S,n} = -p, \quad \left(\frac{\partial U}{\partial n}\right)_{S,V} = \mu$$

Meaning of μ

$$dG = -SdT + Vdp + \mu dn$$

This implies that $G = G(T, p, n)$.

G is extensive, and n is the only extensive variable. Thus,

$$G(T, p, n) = ng(T, p)$$

and

$$\left(\frac{\partial G}{\partial n}\right)_{T,p} = \left[\frac{\partial\{ng(T, p)\}}{\partial n}\right]_{T,p} = g(T, p)$$

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Meaning of μ

$$g(T, p) = \frac{G}{n} = \left(\frac{\partial G}{\partial n} \right)_{T, p}$$

However,

$$dG = -SdT + Vdp + \mu dn$$

gives

$$\mu = \left(\frac{\partial G}{\partial n} \right)_{T, p} = g(T, p) = \frac{G}{n}$$

(molar Gibbs energy)

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Thermodynamic Potentials

$$\mu = G/n$$

can be written as

$$G = \mu n$$

From this equation, we can obtain

$$H = G + TS = \mu n + TS$$

$$A = G - pV = \mu n - pV$$

$$U = G + TS - pV = \mu n + TS - pV$$



Gibbs-Duhem Equation

$$U = \mu n + TS - pV$$

Differentiating both sides,

$$dU = \mu dn + n d\mu + T dS + S dT - p dV - V dp$$

but we know that

$$dU = T dS - p dV + \mu dn$$

Thus,

$$S dT - V dp + n d\mu = 0 \text{ (Gibbs-Duhem equation)}$$

$$d\mu = -S_m dT + V_m dp$$

Two Systems in Contact

$$dG = -SdT + Vdp + \mu dn$$

At constant T and p ,

$$dG = \mu dn$$

For two systems in contact,

$$dG_1 = \mu_1 dn_1$$

$$dG_2 = \mu_2 dn_2$$

The total change in Gibbs energy is

$$dG = dG_1 + dG_2 = \mu_1 dn_1 + \mu_2 dn_2$$

Diffusive Equilibrium

$$dG = dG_1 + dG_2 = \mu_1 dn_1 + \mu_2 dn_2$$

If the two subsystems consist of the same type of particles and can exchange particles,

$$dn_1 + dn_2 = 0$$

Hence,

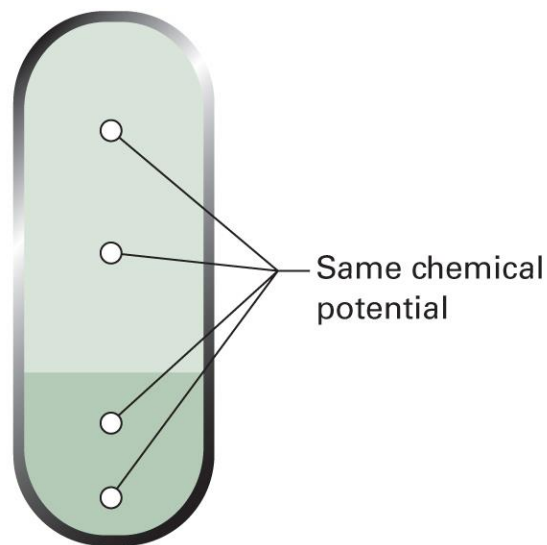
$$dG = \mu_1 dn_1 - \mu_2 dn_1 = (\mu_1 - \mu_2) dn_1$$

At equilibrium, $dG = 0$, and

$$\mu_1 = \mu_2$$

Criteria for Phase Stability

At equilibrium, the **chemical potential** of a substance is the **same** in and throughout every phase in the system



What If $\mu_1 \neq \mu_2$?

$$dG = \mu_1 dn_1 - \mu_2 dn_1 = (\mu_1 - \mu_2) dn_1$$

- If $\mu_1 > \mu_2$, for $dG < 0$ (spontaneous change),

$$dn_1 < 0 \text{ and } dn_2 = -dn_1 > 0,$$

which means that the particles move from system 1 to 2

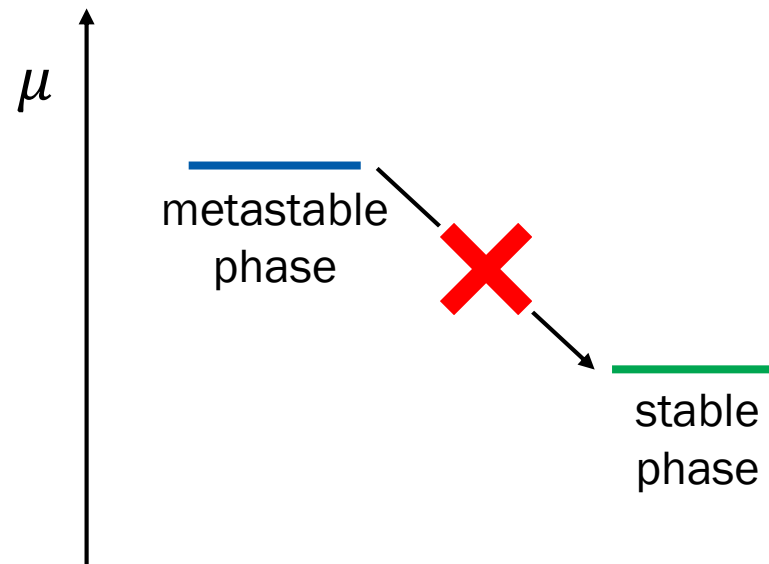
- If $\mu_1 < \mu_2$, for $dG < 0$ (spontaneous change),

$$dn_1 > 0 \text{ and } dn_2 = -dn_1 < 0,$$

which means that the particles move from system 2 to 1

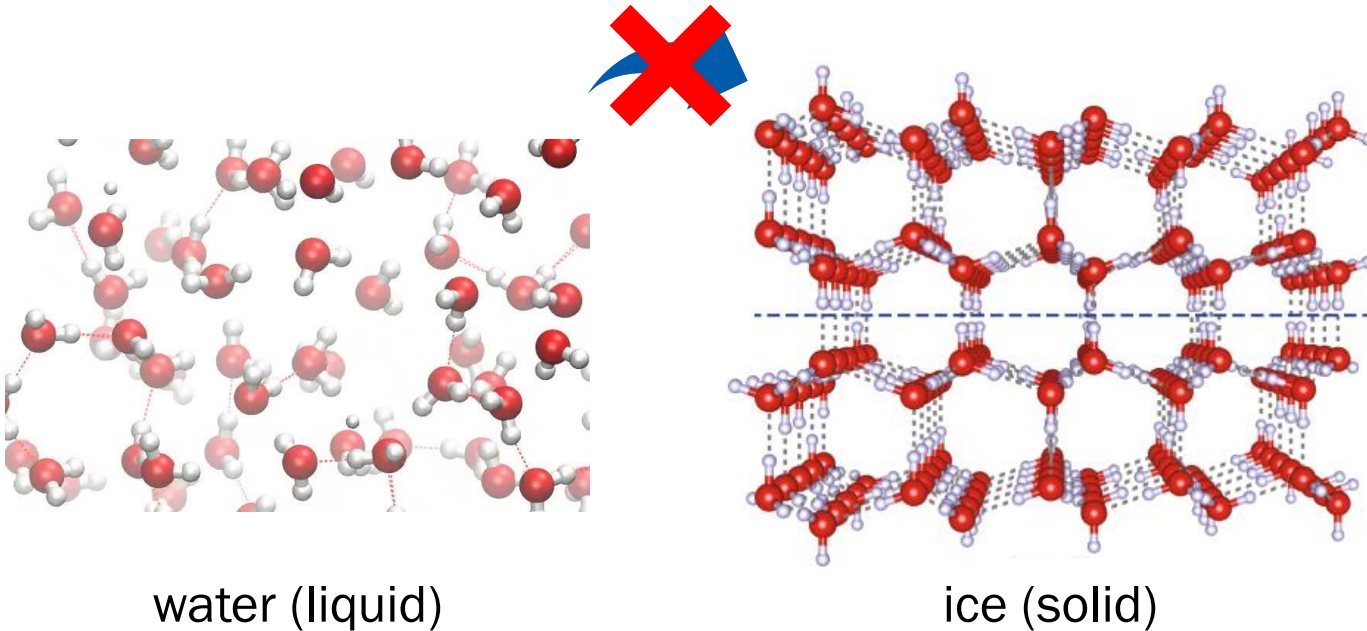
Metastable Phase

Thermodynamically unstable phases that persist because the transition is kinetically hindered



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Supercooling



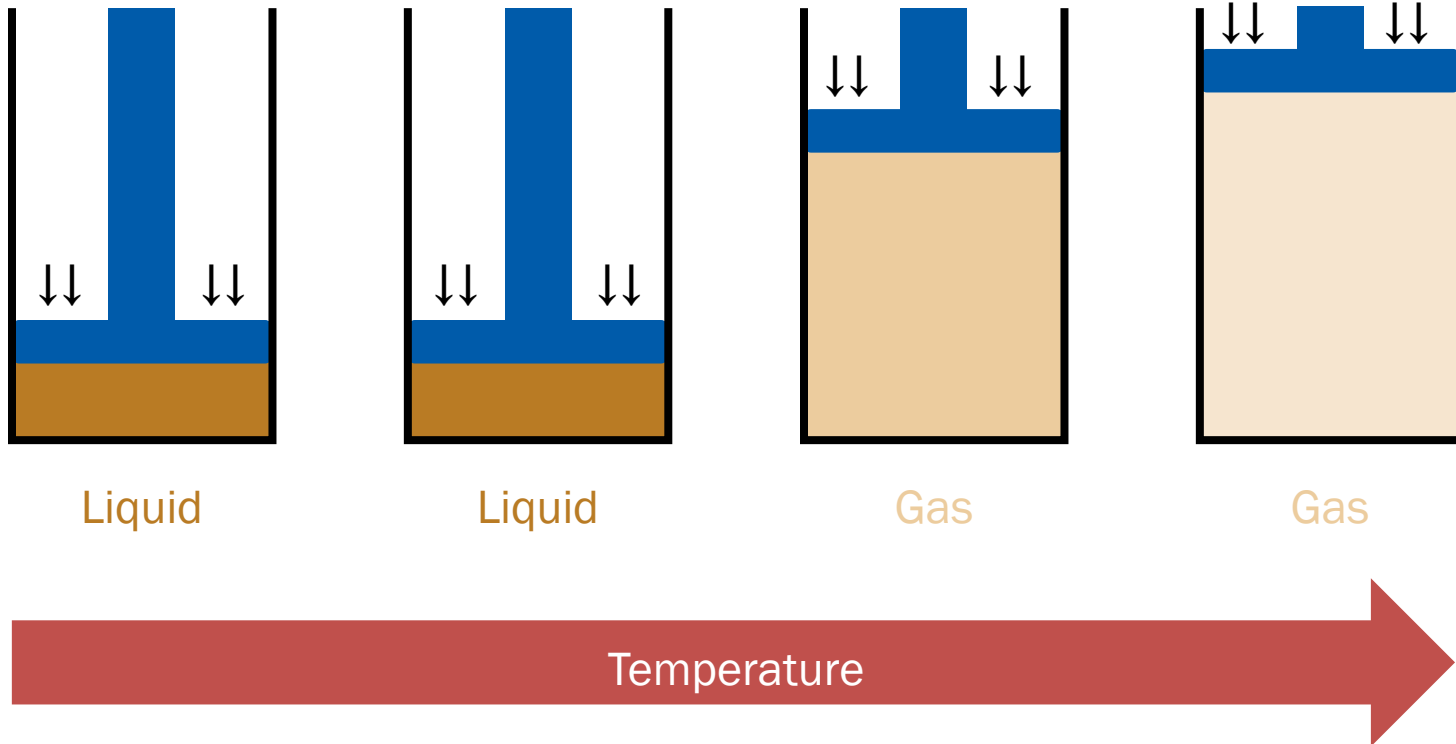
https://www.youtube.com/watch?v=_9N-Y2CyYhM

Let us focus on **phase transitions**

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Liquid-Gas Transition 1

(Isobaric change)

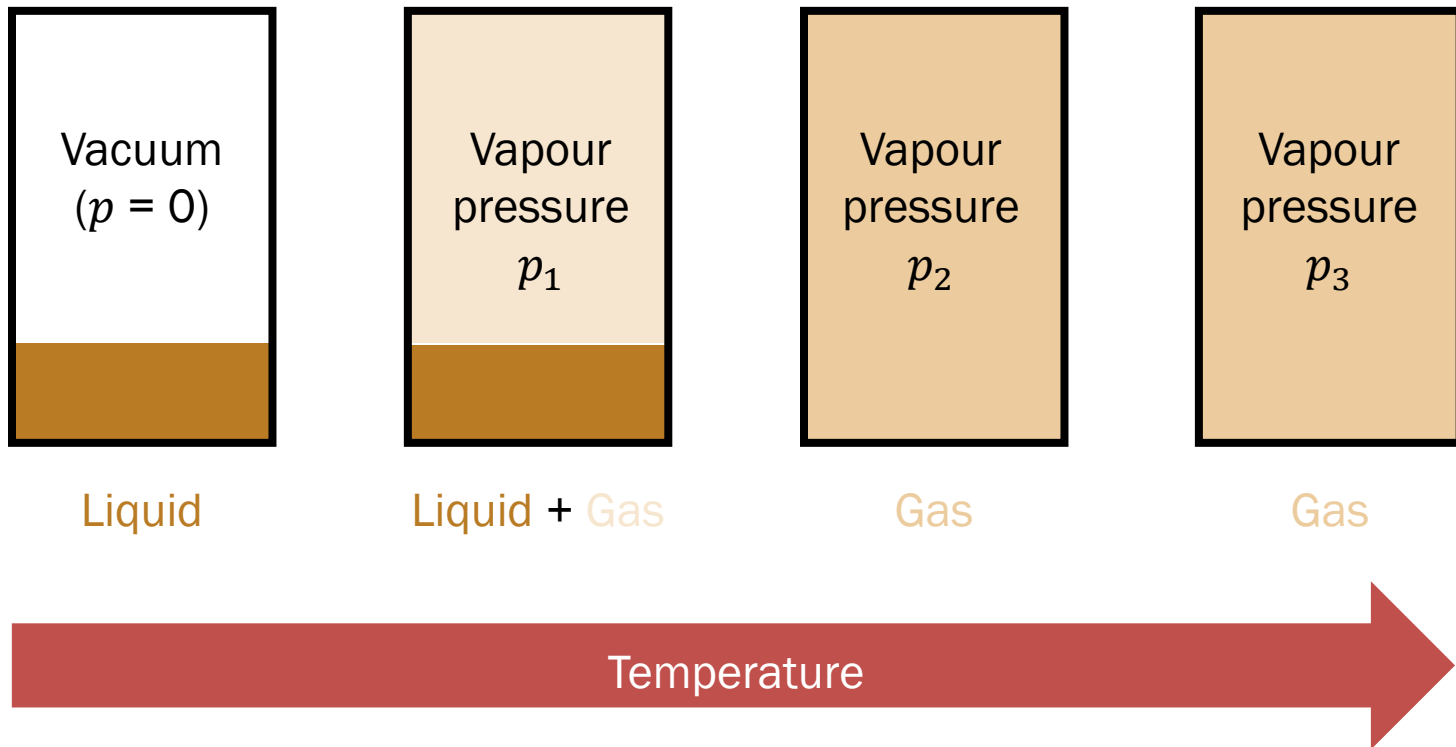


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Liquid-Gas Transition 2

(Isochoric change)

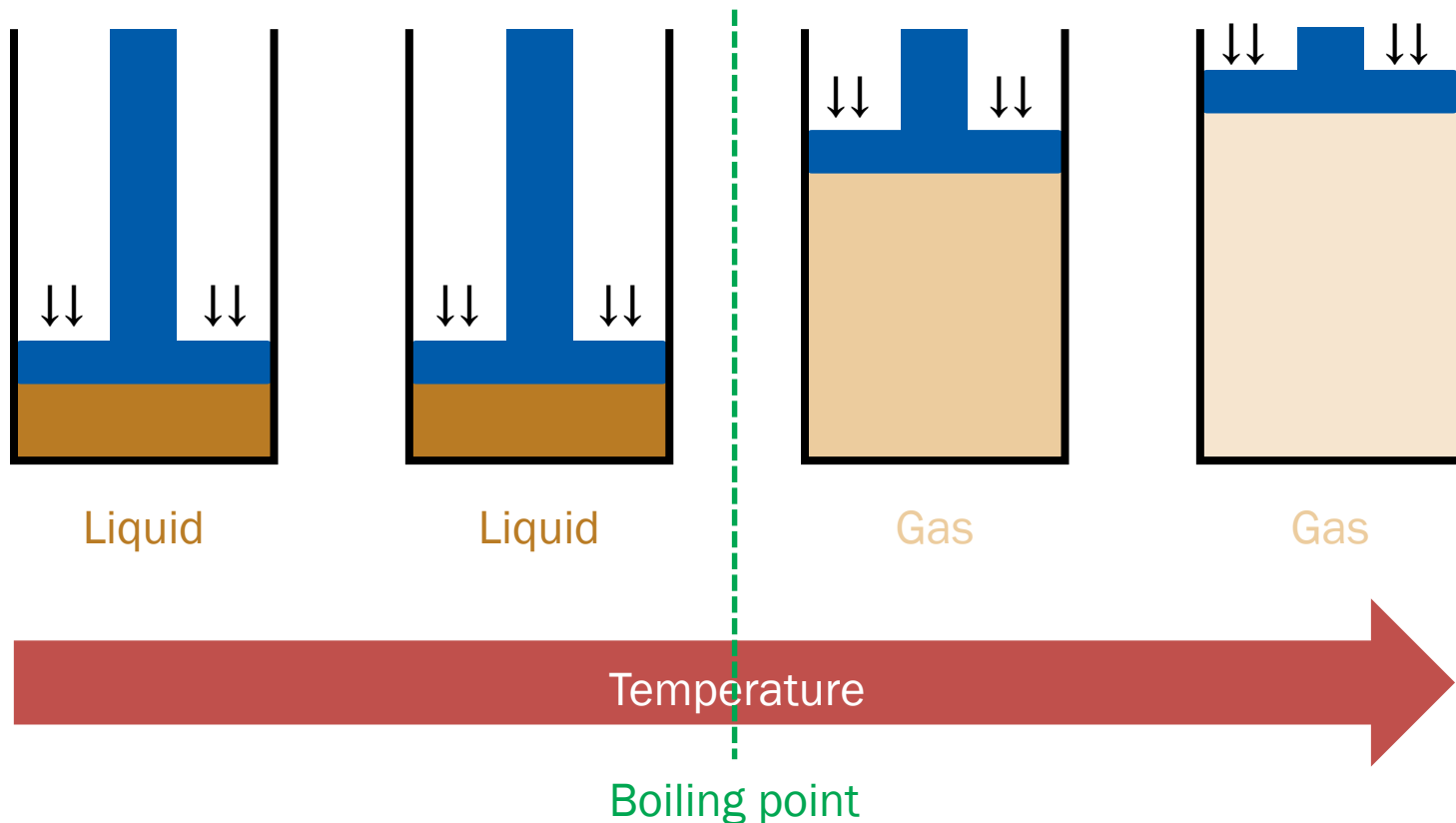
$(p_1 < p_2 < p_3)$



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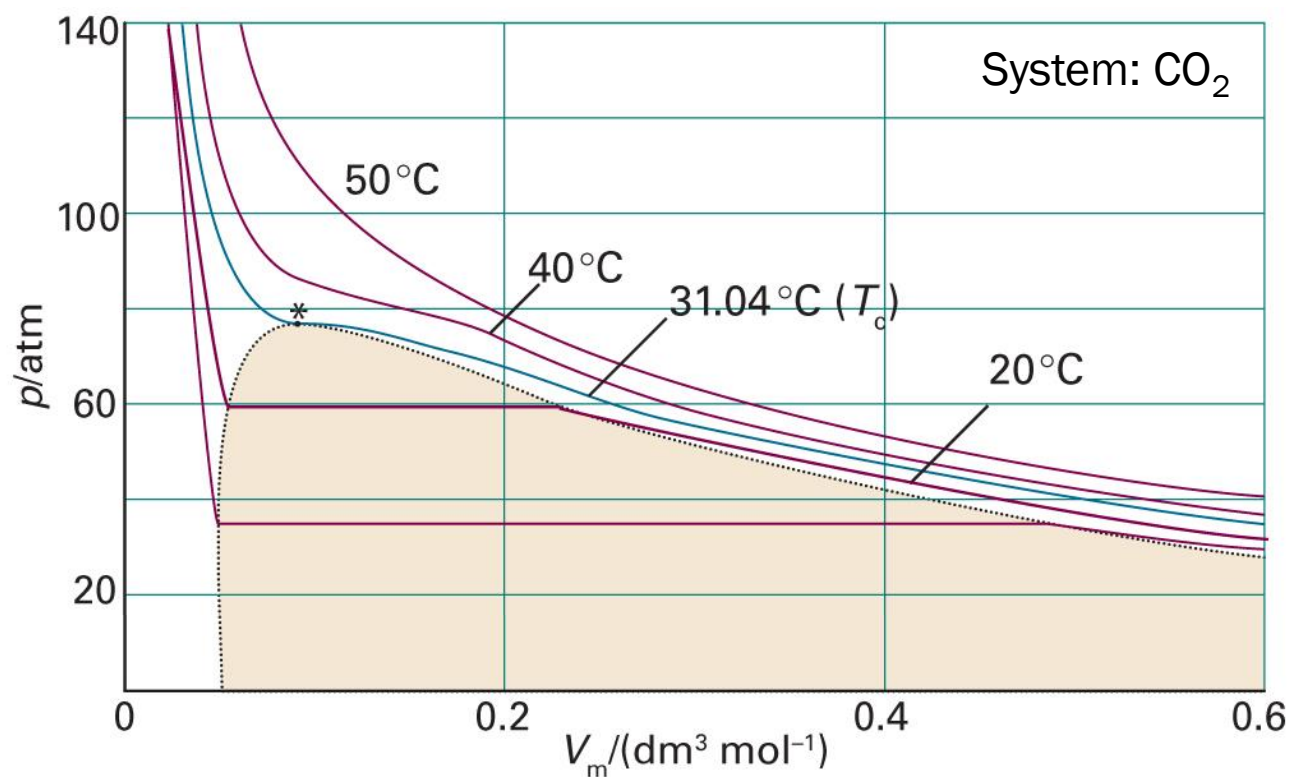
Vapor Pressure and Boiling

(Isobaric change)



Liquid-Gas Transition 3

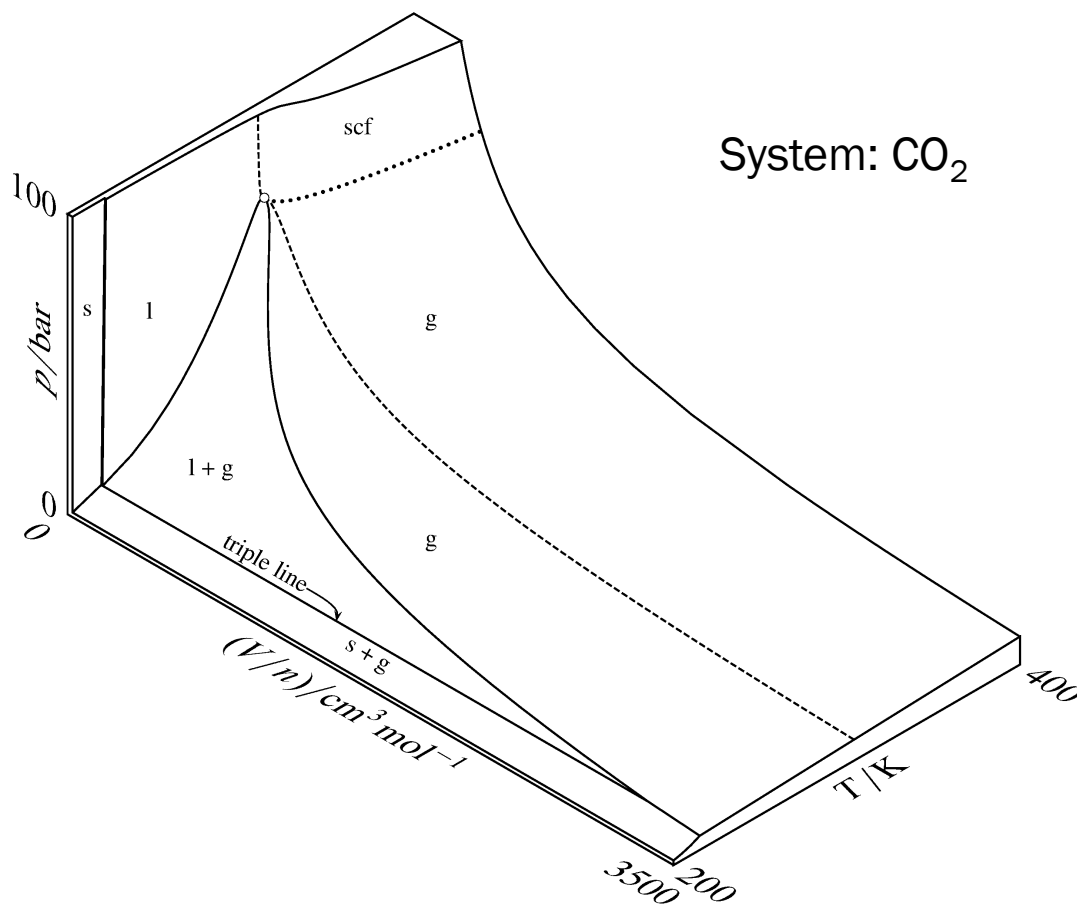
(Isothermal change)



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figures 1C.5)

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3-Dimensional Surface



(Image: <https://chem.libretexts.org/@go/page/20608>)